Quantification: 548_hyp_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	27
interior lacuna count	21
frame-edge lacuna count	6

Interior lacunae only: mean, median and sample SD (n = 21)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2287.24	1100	2322.56	px^2
major_axis_length_px	lacuna shape	74.8952	60.6773	38.2985	px
minor_axis_length_px	lacuna shape	36.8938	27.1347	20.3239	px
aspect_ratio	lacuna shape	2.1879	2.0313	0.7257	unitless
eccentricity	lacuna shape	0.8496	0.8704	0.0939	unitless
solidity	lacuna shape	0.7915	0.7938	0.0868	unitless
roots_count	headline	8.0952	7	4.989	count
ring_length_r30_px	headline	246.238	228	141.27	px
ring_length_r60_px	reported	464.619	451	264.804	px
owned_length_px	ownership-dependent	633.582	414.12	574.696	px
edge_count	ownership-dependent	33.619	22	29.0628	count
mean_edge_length_px	ownership-dependent	17.9458	17.4807	3.3695	px
perimeter_px	normalised	256.659	185.238	168.322	px
ring_area_r30_px2	normalised	7814.33	6495	3744.72	px^2
ring_area_r60_px2	normalised	15865.3	13595	7877.89	px^2
in_frame_fraction_r30	normalised	0.9891	1	0.0346	unitless
in_frame_fraction_r60	normalised	0.9644	1	0.0725	unitless
ring_density_r30	normalised	0.0309074	0.0302568	0.00940961	px^-1
ring_density_r60	normalised	0.0285083	0.0276083	0.0077526	px^-1
roots_per_100px_perimeter	normalised	3.4654	3.2745	1.4566	per 100 px
ring_attached_length_r30_px	network v2	216.714	221	131.403	px
ring_attached_length_r60_px	network v2	339.619	309	209.163	px
ring_length_w_r30_px	network v2	278.685	253.309	160.936	px
ring_length_w_r60_px	network v2	529.293	505.588	305.866	px
sholl_crossings_r10	network v2	5.9048	6	3.3152	count
sholl_crossings_r20	network v2	6	5	3.8079	count
sholl_crossings_r30	network v2	5.8095	5	3.5443	count
loop_count	loops, report only	1.6667	1	2.5755	count
loop_edge_share	loops, report only	0.1071	0.0882	0.1251	share
ring_width_mean_r30_px	width	6.8378	6.845	1.2142	px

Per field
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measure	value	units
analysed_area_px2	990302	px^2
total_skeleton_length_px	25190	px
canalicular_length_density_per_px	0.0254367	px^-1
junction_count	515	count
junction_density_per_px2	0.00052004	px^-2
lacuna_count	27	count

lacunae_per_px2	2.726e-05	px^-2
skeleton_component_count	431	count
mean_component_length_px	58.4455	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0289358	px^-1
field_density_without_flagged_per_px	0.024178	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	541	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.0254367	px^-1
field_length_density_with_unattached_per_px	0.025983	px^-1
loop_count	46	count
loop_edge_share	0.113	share
edge_count	1451	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.469	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	21746	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1331	68.1738	43.3369	1.5731	0.772	0.5571	0.8707	22.9699	562.642	True
2	983	77.6846	19.2927	4.0266	0.9687	0.7097	-0.7276	26.3245	837.443	True
3	1345	60.6773	30.0142	2.0216	0.8691	0.8739	0.6556	34.9123	469.557	False
4	5304	115.818	61.0253	1.8979	0.8499	0.8497	-0.5279	76.0664	899.181	False
5	1100	56.8071	27.1347	2.0935	0.8785	0.8035	0.1617	61.0436	700.865	False
6	575	38.3392	24.0597	1.5935	0.7786	0.7546	0.9307	105.296	722.193	False
7	573	40.0083	19.0937	2.0954	0.8788	0.8788	-1.5045	203.53	964.757	False
8	589	35.3245	25.9374	1.3619	0.6789	0.8158	0.6135	217.115	155.742	False
9	1020	65.1958	29.2967	2.2254	0.8933	0.5543	0.7245	258.288	921.093	False
10	508	48.1299	13.5874	3.5423	0.9593	0.9153	0.8262	258.841	719	False
11	2999	76.2507	53.7089	1.4197	0.7098	0.8844	-0.1093	288.357	25.6549	True
12	8181	176.425	74.8525	2.357	0.9055	0.7394	-0.9469	333.681	384.601	False
13	504	57.8878	15.1903	3.8108	0.965	0.6512	0.9156	299.101	764.925	False
14	772	43.3165	23.8026	1.8198	0.8355	0.8863	1.0039	324.626	799.803	False
15	843	67.1384	19.7863	3.3932	0.9556	0.7938	-0.8751	303.813	882.996	False
16	5780	125.902	64.4212	1.9544	0.8592	0.8211	0.4178	372.596	722.104	False
17	680	44.3797	22.5335	1.9695	0.8615	0.8543	-0.0423	380.915	8.0779	True
18	3628	78.9766	63.2544	1.2486	0.5988	0.8761	0.6151	443.631	464.428	False
19	561	40.1918	19.6867	2.0416	0.8718	0.8738	0.4324	421.501	794.918	False
20	1526	56.8255	39.8642	1.4255	0.7126	0.7592	0.9788	480.14	672.478	False
21	2634	99.9739	44.742	2.2345	0.8943	0.697	1.1806	489.293	203.02	False
22	5796	118.655	71.2141	1.6662	0.7999	0.7445	0.2563	543.606	765.066	False
23	527	38.3444	21.2336	1.8058	0.8327	0.7784	0.8959	666.97	249.159	False
24	2266	83.3012	36.8142	2.2627	0.897	0.8389	0.7457	720.831	996.501	True

25	1380	84.7713	25.484	3.3264	0.9537	0.7908	-1.481	787.319	890.853	False
26	1983	87.7473	31.9283	2.7483	0.9315	0.8015	-1.3962	877.209	988.193	True
27	4886	124.09	61.0897	2.0313	0.8704	0.7632	0.9159	965.955	722.24	False

Network per lacuna

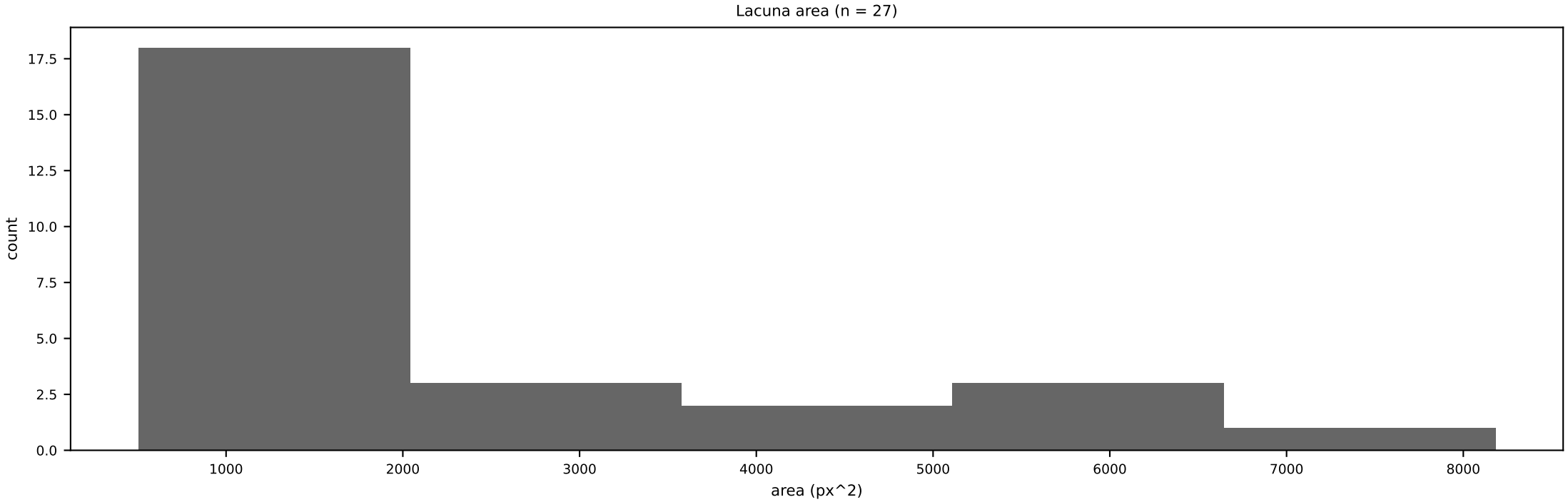
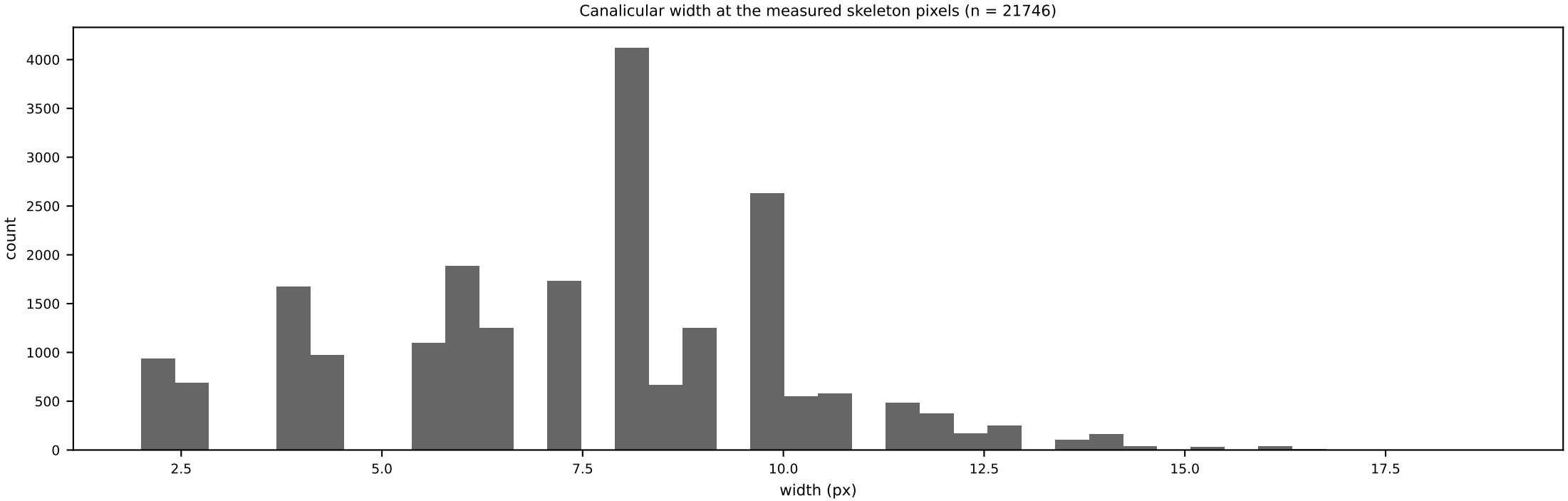
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	6	203	364	599.428	19	31.5489	10.1383
2	6	201	333	486.475	34	14.3081	6.7471
3	4	237	451	414.576	17	24.3868	6.6843
4	9	356	526	499.931	23	21.7361	8.2994
5	8	222	288	236.752	14	16.9109	6.6476
6	7	228	473	414.12	25	16.5648	5.7611
7	6	133	232	167.711	10	16.7711	6.3009
8	7	226	491	359.907	18	19.9948	6.4606
9	12	266	442	317.907	19	16.7319	7.2483
10	5	120	276	187.539	10	18.7539	5.9162
11	5	207	361	944.021	53	17.8117	7.5914
12	24	658	1200	1942.8	103	18.8621	6.7801
13	3	36	36	20.4853	2	10.2426	
14	2	119	137	158.167	12	13.1805	6.9034
15	7	151	354	319.978	19	16.8409	6.4523
16	7	212	362	317.279	19	16.6989	8.2483
17	2	60	135	24.7279	1	24.7279	7.4395
18	12	355	734	638.316	40	15.9579	6.7865
19	2	88	171	1454.55	62	23.4605	2.8258
20	13	284	494	656.085	34	19.2966	7.4281
21	11	423	910	640.328	34	18.8332	7.5163
22	13	395	673	1899.03	96	19.7816	8.5257
23	6	100	405	850.394	40	21.2599	7.0981
24	5	262	440	165.225	9	18.3584	7.2195
25	4	260	521	1520.82	87	17.4807	7.947
26	7	252	529	571.872	26	21.9951	8.3853
27	8	302	581	288.551	22	13.116	6.926

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	299.12	6629	11618	0.7226	0.6609	0.030623	0.0313307	2.0059
2	191.238	5199	8058	0.7613	0.6818	0.0386613	0.0413254	3.1375
3	185.238	6495	12887	0.8654	0.7462	0.0364896	0.0349965	2.1594
4	355.505	11178	20813	0.996	0.8914	0.0318483	0.0252727	2.5316
5	180.894	5735	10481	1	0.8909	0.0387097	0.0274783	4.4225
6	138.711	4371	10506	1	1	0.052162	0.0450219	5.0465
7	108.811	5247	10269	1	0.9129	0.0253478	0.0225923	5.5141
8	130.368	5996	17021	1	1	0.0376918	0.0288467	5.3694
9	247.806	7003	13467	1	1	0.0379837	0.032821	4.8425
10	111.539	4307	9997	1	1	0.0278616	0.0276083	4.4827
11	300.635	6306	12787	0.711	0.6516	0.0328259	0.0282318	1.6631
12	732.927	17463	35109	1	1	0.0376797	0.0341793	3.2745
13	147.581	3054	4424	1	1	0.0117878	0.00813743	2.0328
14	129.095	3933	5278	1	1	0.0302568	0.0259568	1.5492
15	153.238	5367	9216	1	1	0.0281349	0.0384115	4.5681
16	413.245	10735	15456	1	1	0.0197485	0.0234213	1.6939
17	129.154	3191	7077	0.5819	0.5549	0.0188029	0.0190759	1.5485
18	300.433	10348	22315	1	1	0.0343062	0.0328927	3.9942
19	106.569	4649	8339	1	1	0.0189288	0.0205061	1.8767

20	246.25	7091	13595	1	1	0.0400508	0.0363369	5.2792
21	351.747	10779	26637	1	1	0.039243	0.034163	3.1272
22	533.985	13725	25712	1	1	0.0287796	0.0261746	2.4345
23	118.953	5937	17484	1	1	0.0168435	0.023164	5.044
24	256.108	6631	13116	0.7153	0.654	0.0395114	0.0335468	1.9523
25	213.137	8610	18670	1	1	0.0301974	0.0279057	1.8767
26	256.551	6858	14534	0.748	0.6875	0.0367454	0.0363974	2.7285
27	483.813	12078	25496	0.909	0.8104	0.0250041	0.0227879	1.6535

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	170	268	230.894	406.149	8	7	6
2	181	282	227.338	382.877	6	3	5
3	232	269	269.622	504.09	6	5	4
4	301	454	400.363	589.73	8	13	9
5	221	273	252.066	326.764	6	4	5
6	184	322	248.853	528.061	4	3	6
7	121	164	146.225	256.894	2	3	3
8	178	325	253.309	556.073	5	5	8
9	250	309	301.936	505.588	7	7	5
10	100	169	130.912	307.865	2	5	4
11	148	221	241.522	422.103	6	6	3
12	632	1022	753.097	1397.51	16	15	13
13	32	32	36.1421	36.1421	1	0	0
14	99	100	142.267	160.995	5	4	1
15	108	270	178.581	422.203	3	2	3
16	182	249	233.338	398.563	9	7	3
17	50	78	63.2132	145.74	1	1	1
18	280	433	392.321	819.34	7	9	11
19	88	145	102.083	199.167	3	2	2
20	284	494	326.25	571.215	5	6	5
21	302	482	474.718	1040.8	9	9	11
22	382	577	455.132	767.068	9	11	7
23	87	356	115.811	462.061	4	2	4
24	140	232	290.137	492.647	5	5	10
25	249	424	303.007	608.127	6	7	10
26	218	418	286.693	610.571	6	6	9
27	239	263	336.35	656.884	7	7	8

548_hyp_HP_z20c1-2: distributions, pre-validation, px